Problem 5: Quitting Behavior

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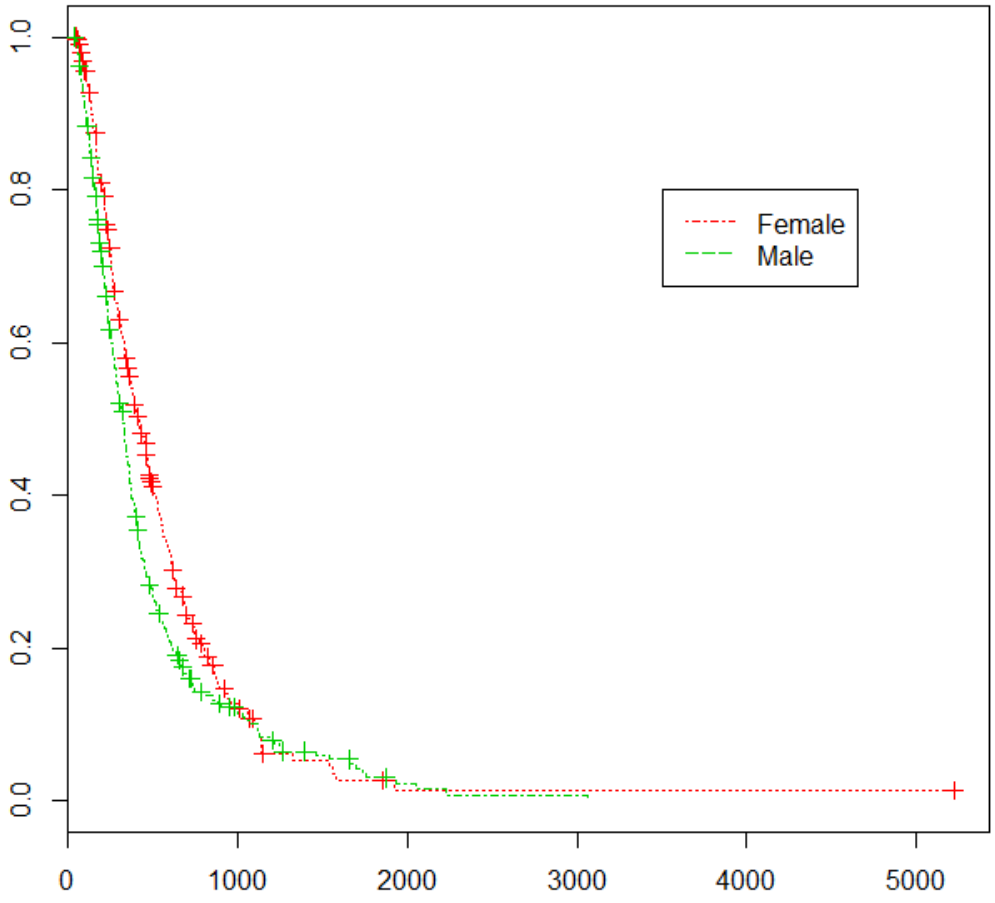
1. KM Estimator



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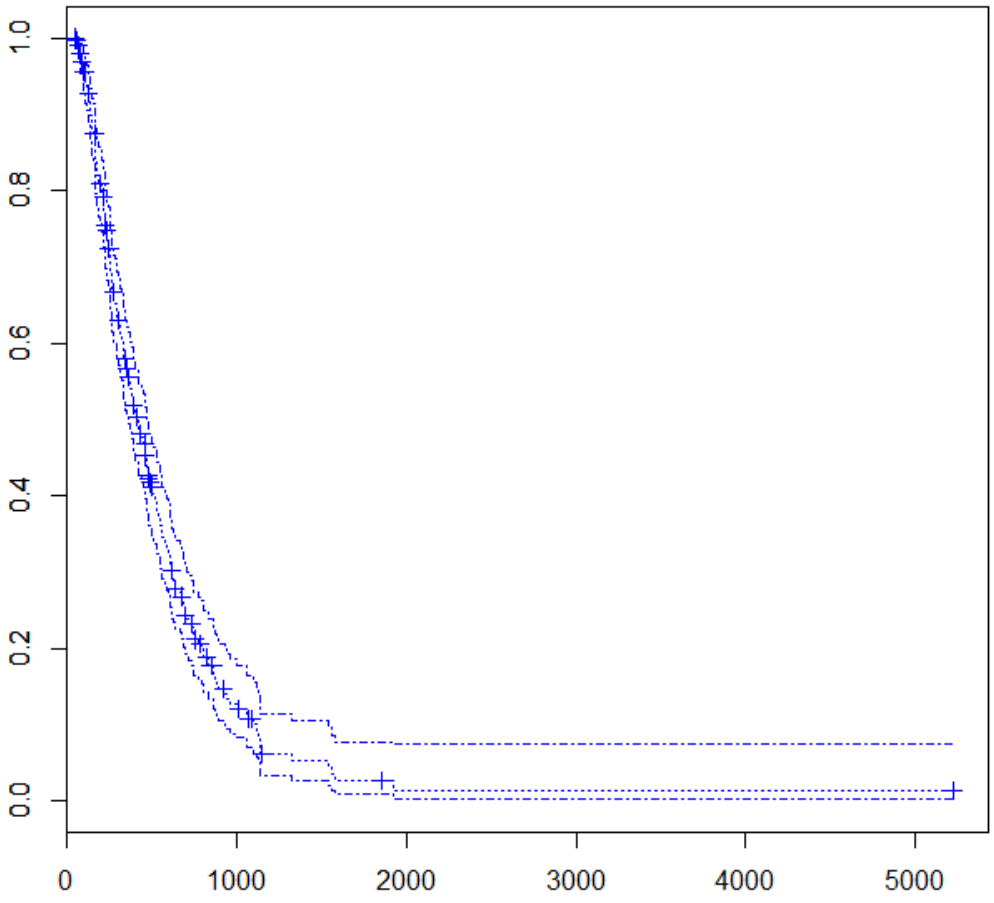

2. Cox proportional hazard model

- $\mathrm{H(t \mid x)} = \exp\left(\beta_1 Sex + \beta_2 Dex + \beta_3 lex + \beta_4 lex^2\right) H_0(t)$
- Hazard Ratio: $\mathrm{H(t \mid x)}/H_0(t)$
- Initial Hazard Rate
- Other covariates shift the initial hazard rate

2. Cox proportional hazard model

	Coeff	S.E	Haz. Ratio	S.E	z	p-value
sex	0.5477	0.0894	1.7293	0.1546	6.127	0.000
dex	-0.1011	0.0067	0.9038	0.0061	-15.066	0.000
lex	-0.8568	0.3056	0.4245	0.1297	-2.804	0.005
lex2	0.0336	0.012	1.0342	0.0124	2.814	0.005